

## Coding & Solution

|                      |                          |
|----------------------|--------------------------|
| <b>Date</b>          | 29 September 2026        |
| <b>Team ID</b>       | 6ab4ce760fc666a751b52130 |
| <b>Project Name</b>  | AI StudyBuddy            |
| <b>Maximum Marks</b> | 5 Marks                  |

### Solution Summary

| Field                                | Details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Repository Link / URL</b>         | (Paste your GitHub repository link here)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Programming Language(s)</b>       | JavaScript (Node.js)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Framework(s) Used</b>             | Express.js v4.22, Mongoose v9.7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Key Features Implemented</b>      | User Registration & Login with JWT (access + refresh tokens). bcrypt password hashing. File upload (.txt, .md, .pdf) via Multer (max 5MB). AI Summary generation via Google Gemini 2.5 Flash. AI Flashcard generation (JSON Q&A). AI MCQ Quiz generation (JSON with options + answers). AI Personalized Study Plan (day-by-day schedule). CRUD operations on study materials. Role-Based Access Control (student/admin). Admin panel — list users, delete users, platform stats. 18 REST API endpoints across 4 route groups. |
| <b>Pending / Incomplete Features</b> | Frontend UI (planned as future enhancement). AI Chatbot for doubt solving (future enhancement). Progress tracking dashboard (future enhancement).                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Setup / Run Instructions</b>      | 1. Clone the repository. 2. Run npm install to install dependencies. 3. Create .env file with PORT, MONGO_URI, JWT_ACCESS_SECRET, JWT_REFRESH_SECRET, GEMINI_API_KEY, NODE_ENV. 4. Run npm start to start the Nodemon development server. 5. Server runs on http://localhost:5000. 6. Test endpoints using Postman.                                                                                                                                                                                                           |

### Code Quality Checklist

| S.No | Criteria                                               | Status (Yes / No) |
|------|--------------------------------------------------------|-------------------|
| 1    | Code is modular and organized into functions / classes | Yes               |
| 2    | Meaningful variable and function names are used        | Yes               |
| 3    | Code includes comments / documentation where necessary | Yes               |
| 4    | Error handling is implemented for critical operations  | Yes               |
| 5    | The application runs without critical errors           | Yes               |
| 6    | Code is committed to a version control repository      | Yes               |

### Additional Notes / Comments

The project follows MVC architecture with clean separation: Models (User.js, Material.js), Views (JSON responses), Controllers (authController, materialController, aiController, adminController). All AI features use a reusable askGemini() utility function. File uploads are cleaned up from disk immediately after text extraction. The express-async-errors package automatically catches all async errors and passes them to the global error handler.